VERIFIED BIBLIOGRAPHIC & EMPIRICAL EXTRACT (House_2011_PNAS)
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Title: Economic and energetic analysis of capturing CO2 from ambient air
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Source: Proceedings of the National Academy of Sciences (PNAS), 108(51), 20428-20433
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Sample: N = Thermodynamic modeling and empirical survey of industrial gas separation & trace-gas systems

EXTRACTED EMPIRICAL METRICS & TABLE COORDINATES:

• Minimum work of separation (Gibbs free energy of mixing): W_min = -R*T0*[ln(y0) + ((1-y0)/y0)*ln(1-y0)]
• Air capture (400-420 ppm CO2): W_min = 20.1 kJ/mol (127 kWh/t CO2) at 50% recovery; 21.8 kJ/mol (138 kWh/t CO2) at 100% recovery
• Point sources: Gas turbine (3.5% CO2) W_min = ~9.8-10.1 kJ/mol; Coal flue gas (12.5% CO2) W_min = ~6.8 kJ/mol; Cement kiln (20%) W_min = ~5.4-5.6 kJ/mol; SMR syngas (45%) W_min = ~2.8-3.2 kJ/mol; Pure streams (>98%) W_min < 0.5 kJ/mol (<3 kWh/t CO2)
• Concentration dilution penalty: 300-fold dilution difference between flue gas (12%) and air (0.04%) increases theoretical W_min by only ~3x, but practical energy increases >10x due to contactor volumes and second-law limits
• Second-law efficiency (eta_II = W_min / W_actual): Industrial point-source amine capture: eta_II = 15-25%; Direct air capture (DAC): eta_II = 3-10% (typically <5%)
• Practical work demand: Existing industrial gas separation systems require >400 kJ of work per mole of CO2 captured from ambient air (requiring carbon-neutral power)
• Sherwood plot scaling: Price/cost scales inversely with dilution: P = A / C; separation constants A: Organics ~$1/kg_i, Metals ~$0.01/kg_i, Gas pollutants ~$0.001/kg_i
• System economics: Total system costs projected on the order of $1,000 / t CO2 based on large-scale trace gas removal empirical analogues

VERBATIM SOURCE QUOTE / TABLE RECORD:

The theoretical minimum work required to separate a stream of air with 400 ppm CO2 into one stream with 200 ppm CO2 and a second stream of highly concentrated (i.e., 99% purity) CO2, all at the same temperature and pressure, is about 20 kJ/(mol CO2). The 300-fold concentration difference of CO2 in flue gas (12%) and air capture (0.04%) causes the minimum work to increase by only about a factor of three... our empirical analyses of operating commercial processes suggest that the energetic and financial costs of capturing CO2 from the air are likely to have been underestimated. Specifically, our analysis of existing gas separation systems suggests that, unless air capture significantly outperforms these systems, it is likely to require more than 400 kJ of work per mole of CO2... We estimate that total system costs of an air capture system will be on the order of $1,000 per tonne of CO2...